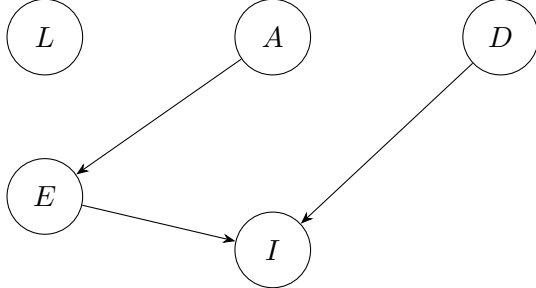


CS 726: Samples questions on Forward and MCMC Sampling

1. For the Bayesian Network H and corresponding potentials below answer the following questions. [Assume the first letter of each node's name as a shortcut for the variable. For example, A is for Age, D for Degree, and so on.]



$$\psi_1(L) = \Pr(L) = \begin{array}{|c|c|c|} \hline \text{NY} & \text{CA} & \text{Other} \\ \hline 0.2 & 0.3 & 0.5 \\ \hline \end{array}$$

$$\psi_2(A) = \Pr(A) = \begin{array}{|c|c|} \hline 0 & 1 \\ \hline 0.6 & 0.4 \\ \hline \end{array}$$

$$\psi_3(D) = \Pr(D) = \begin{array}{|c|c|} \hline \mathbf{0} & \mathbf{1} \\ \hline 0.3 & 0.7 \\ \hline \end{array}$$

$$\psi_4(E, A) = \Pr(E|A) = \begin{array}{|c|c|c|} \hline A \rightarrow & \mathbf{0} & \mathbf{1} \\ \hline \mathbf{0} & 0.9 & 0.3 \\ \hline \mathbf{1} & 0.1 & 0.7 \\ \hline \end{array}$$

$$\Pr(I|D, E) = \Pr(I|D + E) \begin{array}{|c|c|c|c|} \hline D + E \rightarrow & \mathbf{0} & \mathbf{1} & \mathbf{2} \\ \hline \mathbf{0} & 0.9 & 0.6 & 0.2 \\ \hline \mathbf{1} & 0.1 & 0.4 & 0.8 \\ \hline \end{array}$$

- (a) If we perform MCMC sampling on the full network, what is the value of the following transition probabilities

i. $\Pr(L, A, D, E, I = \text{NY}, 0, 1, 1, 0 | L, A, D, E, I = \text{NY}, 0, 0, 1, 0)$..1

$$\frac{1}{5} * \frac{\Pr(D=1) * \Pr(I=0 | D=1, E=1)}{\Pr(D=1) * \Pr(I=0 | D=1, E=1) + \Pr(D=0) * \Pr(I=0 | D=0, E=1)}$$

ii. $\Pr(L, A, D, E, I = \text{NY}, 0, 0, 1, 0 | L, A, D, E, I = \text{NY}, 0, 0, 1, 0)$..2

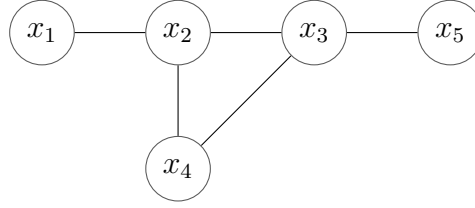
$$\begin{aligned} & \frac{1}{5} \Pr(A = \text{NY}) \\ & + \frac{1}{5} \frac{\Pr(A=0) * \Pr(E=1 | A=0)}{\Pr(A=0) * \Pr(E=1 | A=0) + \Pr(A=1) * \Pr(E=1 | A=1)} \\ & + \frac{1}{5} \frac{\Pr(D=0) * \Pr(I=0 | D=0, E=1)}{\Pr(D=1) * \Pr(I=0 | D=1, E=1) + \Pr(D=0) * \Pr(I=0 | D=0, E=1)} \\ & + \frac{1}{5} \frac{\Pr(E=1 | A=0) * \Pr(I=0 | D=0, E=1)}{\Pr(E=0 | A=0) * \Pr(I=0 | D=0, E=0) + \Pr(E=1 | A=0) * \Pr(I=0 | D=0, E=1)} \\ & + \frac{1}{5} \frac{\Pr(I=0 | D=0, E=1)}{\Pr(I=0 | D=0, E=1) + \Pr(I=1 | D=0, E=1)} \end{aligned}$$

- (b) Now suppose we wish to try sampling for answering the query: $\Pr(I | E = 1)$,

i. For MCMC, what is the transition probability $\Pr(L, A, D, E, I = \text{NY}, 0, 1, 1, 0 | L, A, D, E, I = \text{NY}, 0, 0, 1, 0)$..1

$$\frac{1}{5} * \frac{\Pr(D=1) * \Pr(I=0 | D=1, E=1)}{\Pr(D=1) * \Pr(I=0 | D=1, E=1) + \Pr(D=0) * \Pr(I=0 | D=0, E=1)}$$

2. In the graphical model below, assume binary variables and potentials as $\psi_{1,2}(x_1, x_2)$, $\psi_{234}(x_2, x_3, x_4)$, $\psi_{35}(x_3, x_5)$. If we are doing MCMC sampling and the current sample is $[0, 0, 0, 0, 0]$, what is the probability that the next sample will be $[0, 0, 0, 0, 1]$. Your expression should involve the minimal number of terms from the potentials and should use exact value (0 or 1) in place of x_i .



$$1/5 \frac{\psi_{35}(0,1)}{\psi_{35}(0,1)+\psi_{35}(0,0)}$$

3. Consider doing Gibbs sampling on a 3×3 pairwise grid network with binary labels 0, 1 and all edges having the same potentials of the form $\psi(0,0) = 0.1$, $\psi(1,1) = 0.6$, $\psi(0,1) = 0.2$, $\psi(1,0) = 0.2$. A state X is the assignment of 0/1 labels to the nine variables of the network and we write them in row major order. For example, $X = 111\ 000\ 000$ denotes that the first row has all 1s and the last two rows has all 0s. Work out the following transition probabilities $T(X \rightarrow X')$ for this network.

- (a) $T(000\ 000\ 000 \rightarrow 000\ 010\ 000)$

$$\begin{aligned} T(000\ 000\ 000 \rightarrow 000\ 010\ 000) &= \frac{1}{9} \frac{\psi(0,1)\psi(0,1)\psi(1,0)\psi(1,0)}{\sum_{v=1,0} \psi(0,v)\psi(0,v)\psi(v,0)\psi(v,0)} \\ &= \left(\frac{1}{9}\right) \frac{.2^4}{.1^4 + .2^4} \\ &= 0.10457 \end{aligned}$$

..1

- (b) $T(000\ 000\ 000 \rightarrow 000\ 000\ 000)$

$$\begin{aligned} T(000\ 000\ 000 \rightarrow 000\ 000\ 000) &= \frac{1}{9} \sum_i P(x_i|x_{-i}) \\ &= \left(\frac{1}{9}\right) \left(4 \frac{.1^2}{.1^2 + .2^2} + 4 \frac{.1^3}{.1^3 + .2^3} + \frac{.1^4}{.1^4 + .2^4}\right) \\ &= 0.14480 \end{aligned}$$

..2

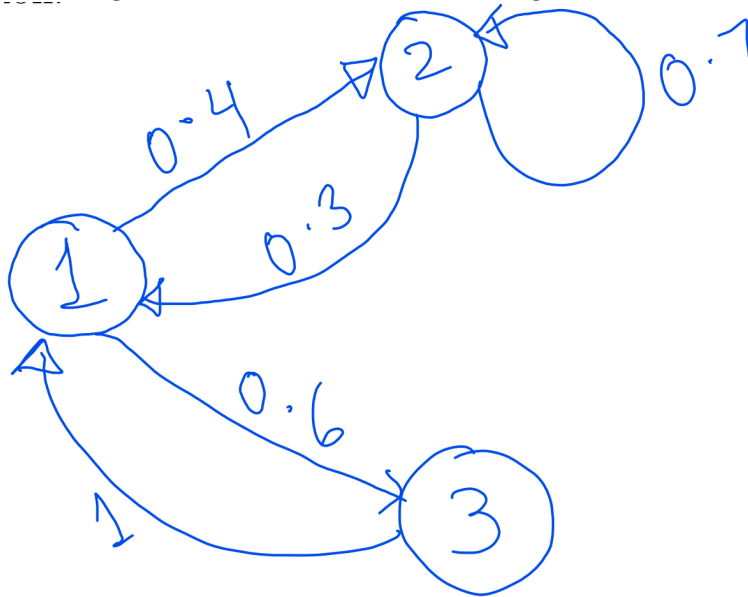
4. Suppose we are computing $T()$ via Gibbs sampling on a Bayesian network with r labels, a maximum of p parents per node and c children per node.

- (a) Write the simplified expression for the conditional probability $\Pr(x_i|x_{-i})$ in a Bayesian network. $\Pr(x_i|x_{-i}) = \frac{\Pr(x_i|Pa(x_i)) \prod_{j \in \text{child}(x_i)} \Pr(x_j|Pa(x_j))}{\sum_{x_i} \Pr(x_i|Pa(x_i)) \prod_{j \in \text{child}(x_i)} \Pr(x_j|Pa(x_j))}$..2

- (b) What is the maximum number of multiplications and additions you need to perform in the above computation? Total number of multiplications = $c(r+1)$
Total number of additions = $r-1$..1

5. Consider a MCMC sampler over the following 3 variables with the transition probabilities

given on the edge. State which of the following is true about this chain with a brief justification.



cation.

- (a) The chain is not regular.
- (b) The chain is regular since any state X' can be reached from any state X in exactly 2 steps.
- (c) The chain is regular since any state X' can be reached from any state X in ≥ 2 steps.
- (d) The chain is regular since any state X' can be reached from any state X in exactly 3 steps.
- (e) The chain is regular since any state X' can be reached from any state X in ≤ 3 steps.
- (f) The chain is regular but none of the reasons above are correct.

The last answer is correct. We need to find k within which all pair transitions are possible in exactly that many steps.. The 1 to 3 transition is not possible in exactly 2 steps Also 3 to 3 is not possible in exactly 3 steps. For $k = 4$ all transitions are possible.

6. In the above graph, calculate the stationary distribution and write the value of $\pi_1 + \pi_2 - \pi_3$ below. Also, upload the derivation. This can be computed easily by solving

$$\begin{aligned}
 \pi_1 &= 0.3\pi_2 + \pi_3 \\
 \pi_2 &= 0.4\pi_1 + 0.7\pi_2 \\
 \pi_3 &= 0.6\pi_1 \\
 \pi_1 + \pi_2 + \pi_3 &= 1
 \end{aligned}$$

The answer is $\pi_1 = \frac{1}{(1+4/3+0.6)}$, etc

..2